

Caesarean section

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Objectives

- Understand the indications, procedure, complications and consequences of caesarean section.
- Introduce the concept of risk management in relation to operative delivery.

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- A caesarean section, also known as **C-section** or **Caesar**, is a surgical procedure in which incisions are made through a mother's abdomen (**laparotomy**) and uterus (**hysterotomy**) to deliver one or more babies, placenta and membranes.

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- It usually indicated when vaginal delivery poses a risk for either the mother or the baby. This operation has been part of human culture since ancient time. Initially it was performed to retrieve the infant from a dead or dying mother.

Incidence:

- Depending on the population and the facilities available the incidence range between 10 and 25%, this difference may be explained by the difference in the demographic and clinical characteristic like maternal age, ethnicity, prior CS, breech presentation...
- In the UK, more than 21% of all deliveries are by C-section.

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- The WHO ranges are between 10-15%. The Scandinavian countries have managed to hold the rate at this level with the same outcome or even better than those with higher rate.
 - More recently it has been suggested that there is maternal and neonatal benefit with caesarean section rates up to 19%



Factors that may contribute to an increase in the rate of C/S:

- In accurate dating of pregnancy.
- Fetal monitoring.
- Macrosomia .
- Maternal request.

Types of C/S:

- ⦿ **Emergency**: immediate threat to mother and fetus.
- ⦿ **Urgent**: maternal or fetal compromise but not immediately life threatening.
- ⦿ **Scheduled**: need early delivery but no maternal or fetal compromise.
- ⦿ **Elective**: at a time to suit the patient and the maternity team.
- ⦿ **Perimortem**: carried out in extremis while the mother is undergoing active resuscitation.
- ⦿ **Post-mortem**: carried out after the death of the mother in order to save the fetus.

Category 1 or emergency cs

- Ideally, CS should be done within the next 30 min.
- examples include abruption, cord prolaps, scar rapture, scalp blood pH < 7.20, and prolonged FHR deceleration below 80.

Category 2 or urgent cs

- The delivery should be completed within 60-75 min. examples include those with FHR abnormalities of concern.

Category 3 or scheduled

- There is concern that continuation of pregnancy is likely to affect the mother or fetus in hours or days.
- This group has a wide range of indications. Examples include failure to progress when CS is planned in next few hours, an IUGR fetus in preterm period with absent EDF but normal CTG, or pre-eclampsia where LFT or RFT are gradually deteriorating and CS is planned within days.
- Timing would vary but some plan should be in place to deliver before further deterioration occurs.

Category 4 or elective cs

- These are cases where there is an indication for CS but there is no urgency. Examples include PP with no active bleeding, breech presentation, previous hysterotomy or vertical incision, past history of repair of VVF.
- Elective CS is generally done around 39 weeks as the incidence of tachypnoea of the newborn is much less after this gestation, however the medical or clinical indication determine time of CS

Indications:

- ⊙ There are many different reasons for performing a delivery by C/S; however absolute indications are very few. Whatever indications, the principle is that ***whenever the risk from the vaginal delivery exceeds that of C/S, operation should be done.***

Current pregnancy events

- ⦿ Significant fetal disease likely to lead to poor outcome like IUGR.
- ⦿ Maternal medical disorder like pre-eclampsia.
- ⦿ Monoamniotic twin.
- ⦿ Complete placenta praevia.
- ⦿ Active primary herpes.
- ⦿ Malpresentation like transverse lie or breech presentation.
- ⦿ Macrosomia (baby is too large).



Past obstetric history:

- Previous classical C/S.
- Secondary pelvic floor or anal sphincter injury.
- Previous severe shoulder dystocia with significant neonatal injury.

Intrapartum events:

- Maternal distress.
- Fetal distress.
- Failed induction.
- Prolonged labor or poor progress.
- Catastrophes like cord prolaps or uterine rapture.
- Placental abnormality like abruption.
- Cephalopelvic disproportion.

Before the procedure:

- **Informed consent:** full informed consent must always be obtained prior to operation. Although it is often difficult to give complete information during emergency situation, mothers must understand what been planned and why.
- All women must be educated in pregnancy about possibility of emergency C/S.

Before the procedure:

- It's important to remember that no adult may give consent for another adult (**although it's important to keep the relative fully informed**), where there is inability to consent (**as may occur in eclamptic fit**), the doctor is expected to act in the patients best interest. When there is significant doubt about decision and capacity to consent, 2 medical opinions are required.



Before the procedure:

- The consent should contain both **risk** and **benefit** of operation. If the patient request tubal ligation, then separate consent for this should be signed.

Preparation:

- All patients being transferred to theater must be in the **left lateral position** with a wedge under the right buttock (to prevent supine hypotension syndrome and fetal distress). Premedication with **antacid** is standard. In theater, the operating table must also be kept in the **left lateral tilt** position until after delivery. **Thromboprophylaxis** should be considered for all patients and prophylactic antibiotic should be given.

Preparation:

- The bladder should be emptied. It is the policy in many units for a catheter to remain in situ peri-operatively and for a defined period thereafter (commonly 24 hours or until the patient is mobile). Although this policy improve patients outcome, but it increase the incidence of infection.

The following should be checked:

- Ask when the patient had anything to eat or drink. Ideally the patient should be fast for at least 8 hours.
- Ask about history of allergy, bleeding tendency, or any medication.

Anesthesia:

- Although the type of anesthesia used is determined by the condition of the mother and the baby, in most cases either a **spinal or epidural** is used, both of which allow the mother to remain awake without any pain. However, **general anesthesia** which makes the mother asleep is sometimes used.

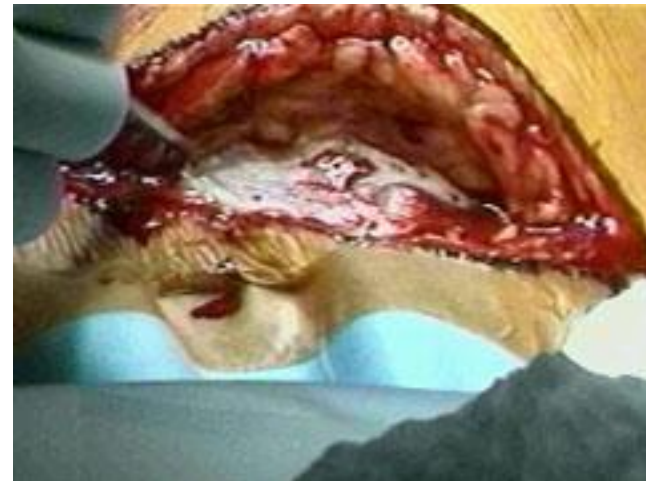
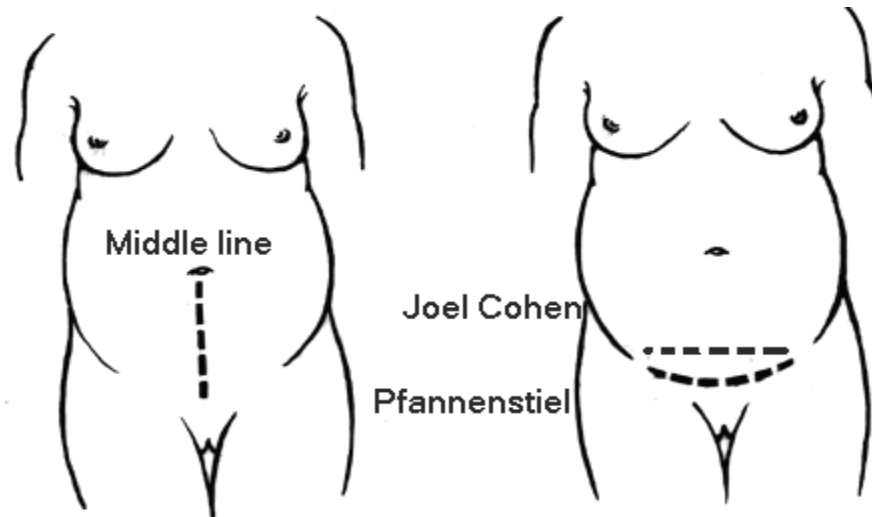
The procedure

■ Skin incision

Two basic skin incisions are used:

- The **transverse** (Pfannenstiel incision) which runs across the pubic hair line, it has the advantage of improved cosmetic results, decreased analgesic requirements and superior wound strength.

Abdominal entry



- The **vertical incision** that runs from umbilicus down to the pubic hair line, which has better surgical access to pelvis, but higher incidence of wound dehiscence (8 fold). It's indicated in extreme maternal obesity, suspected intra-abdominal pathology

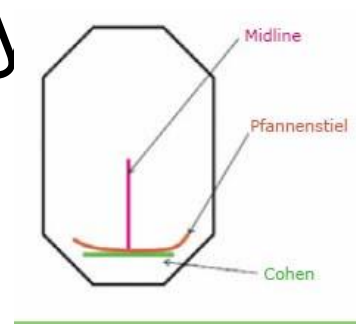
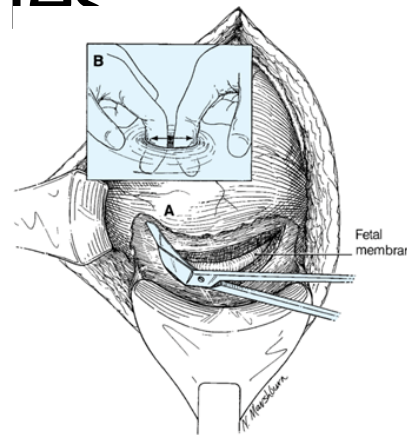
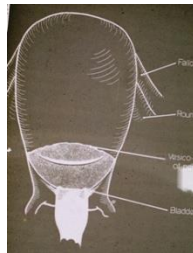


Figure 3. Commonly used skin incisions for caesarean sections.

Uterine incisions:

- The lower uterine incision is used in over 95% of Caesarean deliveries due to ease of repair, reduced blood loss and low incidence of dehiscence or rupture in subsequent pregnancies



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- There are few absolute indication for classical section (which incorporate the upper vertical uterine segment) these include big lower uterine segment fibroid, dens adhesion covering the lower uterine segment, placenta praevia, transverse lie with back inferior, conjoined twins and carcinoma of cervix.



Closure:

- Closure of uterus should be performed in double layer continuous suture; initial suture should be placed just lateral to the **angle** of the incision. If there is any bleeding a **figure-of-eight** sutures can be employed to control bleeding. **No. 2 vicryl** is used for uterine closure. Peritoneal closure is unnecessary.
- Abdominal closure is performed in the anatomical planes with high strength, low reactivity materials, such as **polyglycolic** acid or **polyglactin**.



Abdominal closure

- Is performed in the anatomical planes.
- Interlocking of the sutures should be avoided as this can cause devascularization of the tissue and delayed healing.

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- The skin may be closed using any number and technique. The most common involve surgical staples, subcutical stiches, or tension suture (interrupted or continuous polyglycolic acid, large-bore monofilament polypropylene or nylon)
 - However lower transverse abdominal skin incision closed with subcuticular stich results in less post operative discomfort and are more cosmetically appealing at the 6 week postoperative visit when compared with staples.

Post CS CARE

- After CS, a trained staff member should observe the woman on a one-to-one basis until she has regained airway control, cardiorespiratory stability and is able to communicate
- After recovery from anaesthesia continue observations (respiratory rate, heart rate, blood pressure, pain and sedation) every half hour for 2 hours and hourly thereafter. If observations are not stable, observe more frequently

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- Women who are recovering well and who have no complications can eat or drink when they feel hungry or thirsty
 - The urinary catheter is usually removed next day and the patient advised to move as early as possible.
 - Remove the urinary bladder catheter once a woman is mobile after a regional anaesthetic and not sooner than 12 hours after the last epidural 'top up' dose

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- Discuss the reasons for the CS and implications for the child or future pregnancies before discharge from hospital.
 - Provide information in verbal and printed formats. If the woman prefers, provide this at a later date
 - Offer early discharge (after 24 hours) with follow up at home to women who are recovering well, are afebrile and do not have complications
 - An appointment to see the patient is usually made after 10-14 days.

Delivery after caesarean section:

- The dictum “once C/S, always caesarean section” is not true, up to 70% of women with previous caesarean section can achieve a vaginal delivery.
- A trial of VBAC should be allowed if there is no contraindication, these contraindications are:
- Absolute contraindication like classical C/S.
- Others like malpresentation, macrosomia, multiple pregnancy, prior 2 or more C/S.



The following increase chance of successful trail:

- Use of low uterine segment incision.
- Non recurring indication for prior operation.
- Smooth prior post operative period.

Repeat caesarean section:

- This is performed through old scar, it carry an increased risk of maternal bleeding, thromboembolism, febrile morbidity, prolonged recovery, long term bladder dysfunction, and increased risk of placenta praevia in subsequent pregnancy.

Risk and complications of C/S:

- **Morbidity and mortality:** confidential Enquiries into Maternal Deaths have enabled the risks associated with different methods of delivery to be analyzed; case fatality rate for all C/S is five times that for vaginal delivery.

Intraoperative complications:

- ◉ Bowel damage, especially during a repeat procedure or with severe adhesions.
- ◉ Urinary tract damage, this increased by prolonged labor, repeat C/S, dense adhesion, and vertical extension of the incision.
- ◉ Bleeding, this may be due to damage to uterine vessel, atony, or placenta praevia.
- ◉ Placenta praevia, this increase with increase in number of C/S.
- ◉ Caesarean hysterectomy, the most common indication are uncontrolled bleeding mainly due to placenta covering previous scar, atony and uterine rupture.

Post operative ; infection

- The incidence is between 1-9%
- These include : fever, wound infection, endometritis, bacteraemia, other serious infection including pelvic abscess, septic shock, necrotizing fascitis and pelvic vein thrombophlebitis) and urinary infection



Pulmonary embolism & DVT

- Its difficult to determine the non fatal incidence, because of different time and way of presentation.
- Always be aware about symptoms of DVT and PE.
- Subjected clinical assessment is unreliable, confirmation by objective test is necessary.
- The index of suspicion is high

Psychological sequel

- This is mainly due to delay in returning the baby.
- Partner support is very important
- The **babies** delivered by C/S has higher rate of transient tachypnea of newborn.
- **Repeat C/S** carry an increased risk of abnormally adherent placenta like placenta accrete, antepartum haemorrhage, prolonged labor, uterine rupture and low birth weight.



Thank you